EEN1095 Git Repository and Development Record

Student Name	Adarsh Pramod	**Student ID**	A12944
Project Title	Prediction of flood using digital twin visualization	**Supervisor**	Dr, Ali Intizar
Implementation Period	Week 1 - Week 14	**Version Date**	12/08/2026
Link to the Repository	https://github.com/adarshpramod2-hub/Flood---Prediction-Digital-Twin	**Shared to Supervisor**	No

Development Record:

Weeks 1-2	
Main implementation / development work completed	The first step of the project was to go through the reference materials of a minimum 25 reference in the initial stage and to see if any changes could be made in the project that is currently being done
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The initial stage involved reviewing academic papers and existing research related to flood prediction and digital twin technology.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	To install the python which shall be working as the back end and the cesiumjs which shall be operated in the front-end

Weeks 3-4	
Main implementation / development work completed	Installation of python and the cesiumJS
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The Backend and the frontend applications (python and cesiumJS) were installed and the parameters which is the backbone of this project has been decided I.e the latitude, longitude, area, and the water depth
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	The next development shall be for the raw datas, which shall be found in a few legally authorised sites of the country. A minimum of 500 raw datas are being targeted

Weeks 5-6	
Main implementation / development work completed	To find out the raw datas and see how well it runs.
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The main aim of this weeks work was to findout the datas see how it runs. Currently found few errors which shall be rectifies the next week work report submission
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	The program was failed and the testing is being done rigourasly

Next development steps	To check if the prediction can be made by using predictive datas rather than the available datas.

Weeks 7-8	
Main implementation / development work completed	Implementation of back end coding (Python)
Key files, folders, or components added / updated	Live data Ingestion- To establish a direct link with the raw data to fetch the real time hydrometric sensor metrics Data fault filtering- Detect broken gauges using error handling block Decoupled Output- Converts datas to JSON file seperating the back end math from the front end map.
Evidence of versioning / commits / iterations	The main aim of this weeks work was to bring in the fully developed backend (python) coding .
Testing, debugging, or validation carried out	A successful testing of the code has been carried out multiple times facing few error, which has been corrected
Problems encountered and how they were addressed	1. Package name error 2. Trailing directory 404 error 3. Loop processing latency 4. Server Bot Block
Next development steps	Developing the cesiumJS and the 3D model digital twin, working on the front end.

Weeks 9-10	
Main implementation / development work completed	Development of the front end (CesiumJS) web platform
Key files, folders, or components added / updated	Nil
Evidence of versioning / commits / iterations	During the development of the front end (CesiumJS), iteration and new slight changes were made in the code in order to get the required 3D model of the globe as per the demand of the project.

Testing, debugging, or validation carried out	Multiple number of testings were carried out in the code , several unsuccessful attempts were encountered during the testing of the code leading to trial of other iterations of the code
Problems encountered and how they were addressed	Few problems were encountered during the testing phase such as the blank pages that was seen in the cesium web page, rejection of the request from the server, Specific name of the cities required for the project were not being shown.
Next development steps	The next development is to resume with the front end work for a more refined and structured way of a 3D presentation of the globe showing specific details like the name of the cities, water level of the cites at the moment and chances of flood in the future.
Weeks 11-12	
Main implementation / development work completed	Developed a structured cesiumjs showing a 3D presentation of the globe pointing the cities and showing the current water level in that particular area and predicting the chances of flood by calculating the level of the water bodies, chances of rain and also the intensity of the rain and other factors that decide the intensity of the flood
Key files, folders, or components added / updated	Updated few factors other than the water level, factors that the meteorologists consider when deciding about the flood
Evidence of versioning / commits / iterations	A few factors were also added while working on the front end such as the speed of the wind, drainage rate, soil capacity, that helps in identifying how intese thae affect of flood can be in a particular area.
Testing, debugging, or validation carried out	Regular testings were made to see if the numbers mentioned were constant. In each testing in different time interval the risk of flood keeps changing when there is a change that affects the factor leading to flood, for example at one point if the water level is high the risk of flood is shown high, if checked after few hours if the water level changes the risk or the chances of flood also changes drastically

Problems encountered and how they were addressed	To obtain the real 3D model was coding errors were encountered , which later got rectified.
Next development steps	The next developmet is to add more factors deciding the intensity of flood in particular area to bring in the accurate percentage of the flood.

Weeks 13-14	
Main implementation / development work completed	Completed the front end work of the respective project.
Key files, folders, or components added / updated	An feature was added namely the XAI driver Analysis.
Evidence of versioning / commits / iterations	XAI driver analysis and prediction horizon was added. In the project, these two features holds a vital role in the project. XAI driver analyse the reason behind the prediction and prediction horizon predicts the flood in the coming hours.
Testing, debugging, or validation carried out	Number of testings were made while adding these features
Problems encountered and how they were addressed	NIL
Next development steps	To prepare the portfolio.